IMPLEMENTATION CONTRACT · NOT A TUTORIAL

TRADING AGENT SYSTEM SPECIFICATION

Paper/research MVP for a private Telegram AI trading agent

LOCKED DECISIONS

Mode: PAPER / RESEARCH only. No real-money or testnet orders in MVP.
Universe: BTC, ETH, SOL perps. Timeframes: 15m, 1h, 4h.
Setups (hypotheses): trend_pullback, breakout_retest, sweep_reclaim.
Stack: one Python worker, Postgres, HL public data, Telegram, one LLM.
Handbook v1.0 is frozen knowledge. This spec is frozen architecture.
NO TRADE is the default. The LLM cannot override hard gates.

Version 1.0 · September 2026 · Follow without inventing architecture

Contents

0. Document Control

This is the implementation-ready system specification for a private Telegram-based AI crypto trading research agent. It is an architecture and engineering contract. A coding agent following this document must not invent a different architecture, expand the universe, add setups, introduce queues, microservices, vector databases, or autonomous strategy rewriting.

Field	Value
Document	Trading Agent System Specification v1.0
Status	FROZEN for MVP implementation
Mode	PAPER / RESEARCH only. No live orders. No testnet orders in MVP.
Knowledge layer	Master Trading Intelligence Handbook v1.0 (frozen; do not rewrite)
Venue data	Hyperliquid public mainnet market data (info + optional websocket)
Execution venue MVP	None. Hypothetical fills only.
Later phase	Hyperliquid testnet execution (designed, not built)
Universe	BTC, ETH, SOL perpetual markets
Timeframes	15m, 1h, 4h
Setups	trend_pullback, breakout_retest, sweep_reclaim — hypotheses only

SYSTEM RULE

NO TRADE is the default output. The scanner running is not a reason to emit a trade. A candidate that fails any hard gate is recorded as NO_TRADE and is never sent to the LLM as a TAKE recommendation opportunity.

SOURCE-OF-TRUTH ORDER

1) Live Hyperliquid official docs for venue mechanics and API fields. 2) This specification for software architecture and coded setup parameters. 3) The Master Trading Intelligence Handbook for interpretation, failure modes, and process rules. 4) The agent's own journal statistics for whether an edge exists. Never invert this order.

0.1 What this system is

A single-process research worker that (a) ingests public market data, (b) computes features and coded setups, (c) applies deterministic gates, (d) optionally asks an LLM to review a structured packet, (e) journals every idea, (f) monitors hypothetical outcomes, (g) publishes Telegram alerts, and (h) computes statistics. It does not place real orders in MVP.

0.2 What this system is not

- Not a live trading bot in MVP.
- Not an autonomous strategy rewriter.
- Not a macro forecasting engine.
- Not an order-flow reconstruction engine from OHLCV.
- Not a multi-tenant SaaS.
- Not a Kubernetes or microservice system.

0.3 Operating budget constraint

Target operating cost about $20/month: one small VPS, Postgres on-box, public HL data, Telegram, and a cheap LLM used only after hard gates pass. Design choices below are made to honor that cap.

1. Architecture Overview

1.1 Runtime shape

One Python 3.11+ process named `worker`. Inside it: APScheduler jobs, a Telegram long-poll or webhook listener on a thread, a Postgres connection pool, an HTTP client to Hyperliquid, and an HTTP client to one OpenAI-compatible LLM endpoint. No message bus. No sidecar.

Component	Responsibility	Must not
worker	Schedule, orchestrate pipelines, persist, alert	Place live orders in MVP
Postgres	All durable state	Be replaced by JSON files except local cache
HL info API	Public market data	Be assumed to match testnet
LLM provider	Review structured packets	Compute numbers or invent levels
Telegram	Operator I/O	Be a control plane for strategy edits
Handbook PDF/text	Frozen knowledge retrieved as static excerpts if needed	Be rewritten by the agent

1.2 Process model

Synchronous pipelines per job. Jobs must be idempotent on (job_name, scheduled_ts). A job that is already RUNNING for that key must not start again. Overlap = skip + log, not queue.

1.3 Data flow (happy path)

```
ingest candles/ctx/l2/news/calendar → integrity → features → regime → detect setups
→ cost + geometry + gates → idea row (always) → LLM only if hard gates pass
→ decision TRADE_PAPER | WAIT | NO_TRADE → telegram if alert-worthy → monitor open paper trades
→ on exit: outcome, MFE/MAE, R, classify loss → stats views → learned_rule proposals (never auto-promote)
```

1.4 Modes

Mode	Allowed
research / paper (MVP)	Hypothetical entries, monitoring, alerts, stats. No venue orders.
halted	Ingest + integrity + health only. No new ideas promoted. Monitor existing paper trades to flat.
testnet_exec (NOT MVP)	Later: signed orders on HL testnet. Separate flag, separate keys, separate acceptance.
mainnet_exec	Forbidden until a future spec. Do not implement a hidden path.

2. Repository Structure

Exact tree. Do not add packages outside this tree in MVP.

```
agent/
pyproject.toml | README.md | .env.example | Dockerfile | docker-compose.yml
```

```
handbook/Master_Trading_Intelligence_Handbook.pdf # frozen blob, not parsed as strategy
handbook/EXCERPTS.md # optional human-curated excerpts; no executable rules
src/agent/
__main__.py # entry: load config, start scheduler + telegram
config.py # env + frozen constants
db.py # pool, migrations runner
models.py # SQLAlchemy or raw SQL helpers — pick ONE, use everywhere
timeutil.py # UTC only, candle open/close math
hl_client.py # Hyperliquid info REST (+ optional WS)
ingest/candles.py
ingest/context.py # metaAndAssetCtxs snapshot
ingest/book.py # L2 snapshots
ingest/news.py
ingest/calendar.py
integrity.py
features.py # indicators + structure
regime.py
setups/base.py
setups/trend_pullback.py
setups/breakout_retest.py
setups/sweep_reclaim.py
geometry.py # entry/stop/target/size
costs.py
gates.py # hard + soft
pipeline.py # decision flow orchestrator
llm/packet.py
llm/client.py
llm/schema.py
paper.py # virtual account + fills
monitor.py
outcomes.py
stats.py
learning.py # proposal generation only
telegram/bot.py
telegram/formatters.py
circuit.py
logging_setup.py
versioning.py
migrations/0001_init.sql
tests/ (unit, integration with fixtures, golden packets)
scripts/backfill_candles.py | eval_walkforward.py | report_stats.py
```

SYSTEM RULE

Do not put strategy parameters in prompts. Parameters live in src/agent/config.py and a row in strategy_version. Changing them requires a new strategy_version_id.

3. Modules and Responsibilities

Module	Owns	Inputs	Outputs
hl_client	HTTP to HL /info; retries; timeouts	type+payload	parsed dict; never raises raw timeout unlogged
ingest.candles	Upsert OHLCV	HL candles	candles rows
ingest.context	Mark/oracle/funding/OI/mid	metaAndAssetCtxs	asset_ctx snapshots
ingest.book	L2 snapshot summaries	l2Book	book_snapshots
ingest.news	Headline store	RSS/HTTP	news_items
ingest.calendar	Event windows	CSV/JSON calendar	calendar_events
integrity	Stale/gap/duplicate/jump checks	latest rows	data_quality flags + halt votes
features	TA + swings + z-scores	candles	feature_snapshots
regime	Label + confidence	features + calendar	regime_snapshots
setups.*	Boolean detect + raw levels from code	features+candles	raw_detections
geometry	entry/stop/targets/size	detection+account	geometry block
costs	fees+slip+funding estimate	book+ctx+hold	cost block
gates	Hard/soft pass/fail	full packet minus LLM	gate_results
pipeline	Flow + persistence + dispatch	clock	ideas + decisions
llm.*	Packet build, call, validate JSON	gated candidate	llm_reviews
paper	Virtual fills and equity	TRADE_PAPER	paper_positions
monitor	MFE/MAE/exit	open positions + new bars	updates
stats/learning	Read-only analytics + proposals	journal	reports / learned_rules draft
telegram	Commands + alerts	operator / events	messages
circuit	halt/resume/loss limits	equity + integrity	system_state

4. Environment Variables

All required. Missing required var = process refuse to start. Never log secret values.

Name	Required	Purpose
DATABASE_URL	yes	postgres://user:pass@127.0.0.1:5432/agent
TELEGRAM_BOT_TOKEN	yes	Bot API token
TELEGRAM_ALLOWED_CHAT_IDS	yes	Comma-separated int64. All others ignored.
LLM_BASE_URL	yes	OpenAI-compatible base, e.g. https://api.openai.com/v1
LLM_API_KEY	yes	Bearer token

Name	Required	Purpose
LLM_MODEL	yes	Model id. Frozen into prompt_version notes when changed.
HL_INFO_URL	yes	Default https://api.hyperliquid.xyz/info
HL_WS_URL	no	Default wss://api.hyperliquid.xyz/ws — unused if WS_ENABLED=false
WS_ENABLED	yes	false in MVP unless WS tests pass
AGENT_MODE	yes	paper | halted (testnet_exec rejected in MVP)
PAPER_EQUITY_USD	yes	Default 10000
RISK_FRACTION	yes	Default 0.005 (0.5% equity at stop)
MIN_R_AFTER_COSTS	yes	Default 1.2
MAX_CONCURRENT_PAPER	yes	Default 3
TAKER_FEE_BPS	yes	Default 4.5 (0.045%). Must be overridable; do not silently use maker.
SLIPPAGE_BPS_FLOOR	yes	Default 2.0 plus impact from L2
HOLD_BARS_DEFAULT	yes	Default 12 decision-TF bars for funding estimate
LOG_LEVEL	yes	INFO
TZ	yes	UTC — process timezone must be UTC

SYSTEM RULE

No exchange private key, wallet mnemonic, or mainnet agent key is loaded in MVP. If such an env var is present, the process must refuse to start. Testnet keys belong only to the later phase.

5. Database Schema

PostgreSQL 16. All timestamps timestamptz UTC. Money and prices NUMERIC(38,18) unless noted. Enums as text + CHECK. Migrations in SQL files, applied on boot if pending.

5.1 candles

Primary store of OHLCV. One row per (venue, asset, timeframe, open_time).

```
candles(id bigserial, venue text, asset text, timeframe text, open_time timestamptz,
close_time timestamptz, o numeric, h numeric, l numeric, c numeric, v numeric,
n_trades int, source text, ingested_at timestamptz, UNIQUE(venue,asset,timeframe,open_time))
```

5.2 asset_ctx

```
asset_ctx(id bigserial, venue text, asset text, ts timestamptz,
mid numeric, mark numeric, oracle numeric, funding numeric, premium numeric,
oi numeric, day_ntl_vlm numeric, impact_bid numeric, impact_ask numeric,
prev_day_px numeric, raw jsonb, UNIQUE(venue,asset,ts))
```

5.3 book_snapshots

Store summary plus raw top levels. Do not store full infinite history of every level — keep 7 days.

```
book_snapshots(id bigserial, venue text, asset text, ts timestamptz,
bid1 numeric, ask1 numeric, bid1_sz numeric, ask1_sz numeric, spread numeric, spread_bps numeric,
bid_sz_5 numeric, ask_sz_5 numeric, imbalance_5 numeric,
```

```
notional_to_10bps numeric, raw_top jsonb, UNIQUE(venue,asset,ts))
```

5.4 news_items / calendar_events

```
news_items(id bigserial, source text, external_id text, ts timestamptz, title text,
url text, body text, assets text[], raw jsonb, UNIQUE(source,external_id))
calendar_events(id bigserial, ts_start timestamptz, ts_end timestamptz, name text,
impact text CHECK (impact in ('low','medium','high')), assets text[], source text)
```

5.5 feature_snapshots / regime_snapshots

```
feature_snapshots(id bigserial, asset text, timeframe text, open_time timestamptz,
features jsonb NOT NULL, computed_at timestamptz, UNIQUE(asset,timeframe,open_time))
regime_snapshots(id bigserial, asset text, timeframe text, open_time timestamptz,
label text, secondary text[], confidence numeric, features_used jsonb,
UNIQUE(asset,timeframe,open_time))
```

5.6 ideas (journal spine)

One row per evaluated candidate-or-scan-cell. A scan that finds no setup still writes a lightweight scan_event; a detected setup always writes an idea even if NO_TRADE.

```
ideas(id uuid primary key, created_at timestamptz, asset text, timeframe text,
direction text CHECK (direction in ('long','short','none')),
setup_id text, strategy_version_id text, prompt_version_id text,
bar_open_time timestamptz, decision text, decision_reason text[],
gates jsonb, geometry jsonb, costs jsonb, features jsonb, regime jsonb,
ctx jsonb, book jsonb, news jsonb, calendar jsonb, hist_cell jsonb,
llm_review jsonb, packet_hash text, data_quality jsonb, confidence numeric)
```

5.7 paper_positions / paper_fills / paper_equity

```
paper_positions(id uuid, idea_id uuid, asset, direction, tf, status,
entry, stop, targets jsonb, size, notional, risk_cash, opened_at, closed_at,
exit_px, exit_reason, realized_r, pnl_usd, fees_usd, funding_usd, slip_usd,
mfe_r, mae_r, mfe_px, mae_px, bars_held, outcome_class, counterfactuals jsonb)
paper_fills(id, position_id, ts, side, px, sz, kind, note)
paper_equity(ts, equity, open_risk, drawdown_from_peak)
```

5.8 llm_reviews / learned_rules / strategy_version / prompt_version / system_state / job_runs / audit_log

```
llm_reviews(id, idea_id, request jsonb, response_raw text, response_json jsonb,
valid boolean, error text, model text, latency_ms, created_at)
learned_rules(id, rule_key, definition jsonb, setup_id, regime, n, mean_r,
ci_low, ci_high, validation_period, strategy_version_id, status,
CHECK (status in ('proposed','rejected','promoted','expired')))
strategy_version(id text pk, created_at, params jsonb, code_git_sha, notes, frozen boolean)
prompt_version(id text pk, created_at, template_hash, model, notes)
system_state(key text pk, value jsonb, updated_at) -- mode, halt reasons, peak equity
job_runs(id, job_name, scheduled_for, started_at, finished_at, status, error, stats jsonb)
audit_log(id, ts, actor, action, payload jsonb) -- telegram commands, halt, resume
```

5.9 Indexes

- candles: UNIQUE + (asset, timeframe, open_time DESC)

- asset_ctx: (asset, ts DESC)
- book_snapshots: (asset, ts DESC)
- feature_snapshots: UNIQUE + (asset, timeframe, open_time DESC)
- regime_snapshots: (asset, timeframe, open_time DESC)
- ideas: (created_at DESC), (asset, setup_id, decision, created_at DESC), (packet_hash)
- paper_positions: (status), (idea_id), (asset, status)
- news_items: (ts DESC), GIN(assets)
- calendar_events: (ts_start, ts_end)
- job_runs: UNIQUE(job_name, scheduled_for)
- learned_rules: (status, setup_id, regime)

6. Market-Data Ingestion

6.1 Venue endpoints (verify against live docs)

Base: POST https://api.hyperliquid.xyz/info with JSON body. Timeouts: 8s connect, 15s read. Retries: 3 attempts, exponential backoff 0.5/1/2s, only on 429/5xx/timeout. Weight-aware: do not burst.

Call	Body	Use
candleSnapshot	{type, req:{coin, interval, startTime, endTime}}	OHLCV backfill and catch-up
metaAndAssetCtxs	{type}	mid, mark, oracle, funding, OI, premium, impactPxs, dayNtlVlm
l2Book	{type, coin}	top-of-book + depth summary; docs: at most 20 levels/side
predictedFundings	{type}	optional; store if present; treat as estimate
fundingHistory	{type, coin, startTime}	backfill hourly funding for stats

API CONTRACT NOTE

Official docs state candleSnapshot returns at most the most recent 5000 candles per market/interval. l2Book response shape is {coin, time, levels:[bids, asks]} with px/sz/n. assetCtxs align by index to meta.universe. If a live payload field is missing, store null and flag data_quality; do not invent.

6.2 Poll cadence (MVP, WS off)

Job	Interval	Action
ctx_poll	15s	One metaAndAssetCtxs; upsert BTC/ETH/SOL
book_poll	15s	l2Book for each of 3 assets
candle_15m	at :00/:15/:30/:45 + 8s	Fetch last 5 bars 15m all assets
candle_1h	at minute 0 + 8s	Fetch last 5 bars 1h
candle_4h	at 0/4/8/12/16/20h + 8s	Fetch last 5 bars 4h
funding_hist	1h	Append latest fundingHistory point
news_poll	5 min	RSS pull
calendar_refresh	6h	Reload calendar file
scan_15m	after 15m candle job	pipeline for 15m

Job	Interval	Action
scan_1h	after 1h candle job	pipeline for 1h
scan_4h	after 4h candle job	pipeline for 4h
monitor_open	15s	Update MFE/MAE; check exits on last/mark
integrity	30s	Staleness + gaps
equity_snap	1 min	Paper mark-to-market
daily_stats	00:05 UTC	Recompute cells

SYSTEM RULE

Scans run only on closed candles. Signal at bar T close is eligible at T close + 8s ingest, never on a forming bar. This is a lookahead-prevention SYSTEM RULE.

6.3 Candle storage rules

- Timezone: UTC. 1h open = epoch divisible by 3600s. 4h open = epoch divisible by 14400s. 15m similarly.
- Upsert on unique key. If incoming H/L/C differs from stored closed bar by more than 0.01%, keep first closed value, write audit anomaly; do not silently rewrite history after 2x interval age.
- v is base-asset volume as returned. Do not convert unless you also store the conversion explicitly.
- Backfill on first boot: 15m last 2000 bars; 1h last 3000; 4h last 2000 — clipped by the 5000-candle API cap.
- Missing bar inside a range = integrity gap. Do not interpolate OHLC.

6.4 L1/L2 handling

L1 derived from L2: bid1, ask1, mid=(bid1+ask1)/2 if both exist. Spread_bps = 1e4*(ask1-bid1)/mid. imbalance_5 = (bid_sz_top5 - ask_sz_top5)/(sum). notional_to_10bps = cumulative bid+ask notional within 10 bps of mid. Used only for cost and liquidity gates. Never as a directional trigger in MVP.

FAILURE / MISUSE

Do not infer hidden stops or liquidation clusters from L2. Do not treat imbalance as a setup.

6.5 Funding and OI

From assetCtx: funding is the current hourly rate as a decimal (docs commonly show 0.0000125 = 0.00125%/hour). Store as returned. OI is in coin units per docs; also store oi_notional = oi * mark. Delta OI computed vs snapshot ~1h ago and vs ~24h ago. Predicted funding stored separately if endpoint exists.

6.6 News collection

MVP sources: two or more public RSS feeds (e.g. venue status + one crypto news RSS). Parse title, time, url. Tag assets by ticker token match on BTC/ETH/SOL/Bitcoin/Ethereum. No scraping of authenticated Twitter. No LLM summarization of the entire web. News is a flag, not a cause.

6.7 Macro / calendar

Ship a checked-in calendar JSON for high-impact windows: FOMC, CPI, PCE, NFP, plus optional crypto events (known unlocks only if dated in that file). Operator can add rows via SQL or file. A high-impact event marks EVENT_HIGH for [ts_start - 30m, ts_end + 15m] globally unless assets[] is specific.

7. Feature Engine

All features computed in code from stored candles + ctx + book. The LLM never computes them. Library: pandas + pandas-ta or ta. Library choice is frozen in strategy_version.params.ta_lib.

7.1 Required feature vector (per asset, per TF, per closed bar)

Key	Definition
ema_20, ema_50	EMA of close, Wilder/standard EMA α=2/(n+1), min periods = n
sma_20	SMA close 20
atr_14	Wilder ATR 14
atr_pct_100	atr_14 / median(atr_14, 100)
adx_14, plus_di, minus_di	Wilder ADX 14
rsi_14	Wilder RSI 14
vol_sma_20	SMA of v, 20
vol_ratio	v / vol_sma_20
ret_1, ret_12	close-to-close log return 1 bar and 12 bars
z_close_20	(close - sma_20) / stdev(close,20)
swing_high, swing_low	N=2 pivot: high with 2 lower highs each side
last_swing_high_px/t, last_swing_low_px/t	most recent confirmed pivots
grammar	HH_HL | LH_LL | MIXED from last 3 confirmed swings
pdh, pdl	previous UTC day high/low from 15m or 1h store
equal_high, equal_low	two swings within 0.15*atr of each other
dist_ema20_atr	(close-ema_20)/atr_14
spread_bps, imbalance_5, notional_to_10bps	from latest book
funding, funding_z_168	current hourly funding; z vs last 168 hours
oi, oi_chg_1h, oi_chg_24h	from ctx
mark, oracle, basis_bps	1e4*(mark-oracle)/oracle

SYSTEM RULE

Warmup: do not emit features until 120 closed bars exist on that TF. Below that: data_quality.warmup_insufficient=true and NO_TRADE.

7.2 Regime classifier

Primary regime TF is the same TF as the setup under evaluation. Additionally compute a 4h regime used as a gate for 15m/1h setups.

Label	Coded condition (all must hold)
TREND_UP	ema_20 > ema_50 AND close > ema_20 AND grammar=HH_HL AND adx_14 >= 18
TREND_DOWN	ema_20 < ema_50 AND close < ema_20 AND grammar=LH_LL AND adx_14 >= 18

Label	Coded condition (all must hold)
RANGE	adx_14 < 18 AND abs(z_close_20) < 1.8
BREAKOUT_CLIMATE	atr_pct_100 rising vs 20-bar ago AND vol_ratio > 1.2 AND not RANGE
HIGH_VOL	atr_pct_100 >= 1.6
LOW_VOL	atr_pct_100 <= 0.7
PANIC	atr_pct_100 >= 2.5 OR (high-low of last 3 bars) >= 4*atr_14
EVENT_HIGH	calendar high-impact window active
UNKNOWN	warmup missing OR integrity fail OR conflicting primary labels

Output: primary {TREND_UP, TREND_DOWN, RANGE, UNKNOWN} plus secondary flags HIGH_VOL, LOW_VOL, PANIC, EVENT_HIGH, BREAKOUT_CLIMATE. confidence = 0.3 if ADX in [16,20] else 0.7 if a primary matches else 0.2. UNKNOWN if PANIC xor integrity fail dominates.

SYSTEM RULE

Compatible regimes: trend_pullback needs TREND_UP (long) or TREND_DOWN (short) and not PANIC and not EVENT_HIGH. breakout_retest needs BREAKOUT_CLIMATE or (not RANGE and not PANIC); reject EVENT_HIGH. sweep_reclaim needs RANGE or TREND_* pullback-like; reject PANIC and EVENT_HIGH. Mismatch → hard NO_TRADE.

8. Setup Detectors (Hypotheses)

Each detector returns either null or a structured detection. Detectors do not size, do not call the LLM, and do not look at future bars. Parameters below are v1 and live in strategy_version.params.

HYPOTHESIS

These three setups are research hypotheses. They are not claimed to have positive expectancy. Promotion to 'trusted' requires the evaluation protocol in Part 16.

8.1 trend_pullback

Field	Rule
Preconditions	Primary TREND_UP or TREND_DOWN; not PANIC; not EVENT_HIGH; confidence>=0.5
Long location	TREND_UP AND low of last 3 bars touched ema_20 band (ema_20 ± 0.25*atr) AND last_swing_low still intact
Short location	Mirror
Trigger	Closed bar rejects the band: long close > ema_20 AND close > prior close
Entry	Next-bar marketable limit at current close (paper: fill = close + slip_model on long)
Stop	min(low of pullback bars, last_swing_low) - 0.25*atr [long]
Target	First of: +1.8R, last_swing_high (if R after costs >= MIN_R)
Invalid pre-entry	Close beyond stop level before entry; grammar flips to MIXED/LH_LL
NO_TRADE extra	stop distance < 0.4*atr; or stop distance > 2.5*atr; or R after costs < MIN_R

8.2 breakout_retest

Field	Rule
Boundary	max(high, 20 bars) for long; min(low, 20) for short — excluding current bar
Break	Prior bar closed beyond boundary by >= 0.10*atr
Retest	Within 12 bars after break, a bar wicks back to boundary ± 0.20*atr and closes back on break side
Entry	Close of the retest confirmation bar, + slip_model
Stop	Beyond retest extreme + 0.20*atr
Target	1.8R
NO_TRADE extra	Break already extended > 2.0*atr from boundary at trigger; RANGE primary; PANIC; EVENT_HIGH

8.3 sweep_reclaim

Field	Rule
Pool	equal_high/equal_low or pdh/pdl
Sweep	A bar trades through pool by >= 0.10*atr
Reclaim	Same bar or next <= 3 bars closes back through pool on the origin side
Long example	Sweep below pdl/equal_low then close back above that level
Stop	Sweep extreme - 0.15*atr
Target	min(1.5R, range mid if defined)
NO_TRADE extra	No reclaim, only a wick; PANIC; EVENT_HIGH; stop < 0.35*atr

SYSTEM RULE

If two setups fire on the same asset+TF+bar in opposite directions: NO_TRADE (conflict). If two fire in the same direction: keep the one with higher planned R after costs; journal the other as NO_TRADE reason=dominated.

9. Geometry, Cost Model, and Hard Gates

9.1 Virtual account

paper_equity starts at PAPER_EQUITY_USD. Risk cash per new idea = RISK_FRACTION * current_equity. size = risk_cash / (abs(entry-stop) + per_unit_exit_cost_estimate). Notional = size * entry. If notional > 5% of 24h dayNtlVlm or > 30% of notional_to_10bps, fail liquidity gate.

9.2 Cost model (always applied before R)

```
fee_round_trip = 2 * TAKER_FEE_BPS / 10000 * notional
slip_bps = SLIPPAGE_BPS_FLOOR + impact_bps
impact_bps = 0 if size_notional <= notional_to_10bps else min(25, 10 * size_notional /
max(notional_to_10bps,1e-9))
funding_est = abs(funding) * HOLD_BARS_DEFAULT * (bar_hours) * notional
bar_hours = 0.25 | 1 | 4 according to TF
cost_R = (fee_round_trip + slip_cost_rt + funding_est) / risk_cash
planned_R_after_costs = raw_R - cost_R
```

Paper fill price: long entry = min(ask1, close) + slip; if book missing, close * (1+slip_bps/1e4). Stop hypothetical fill = stop * (1 - 0.5*slip_bps/1e4) for longs (worse). Do not assume perfect stop prices.

9.3 Hard gates (any fail → decision=NO_TRADE, skip LLM TAKE path)

Gate	Fail if
data_valid	integrity fail, warmup fail, stale ctx>60s, stale book>60s, missing mark
regime_ok	UNKNOWN, PANIC, EVENT_HIGH, or setup/regime mismatch
setup_complete	missing entry or stop or target
invalidation_clear	stop on wrong side; stop == entry
min_r	planned_R_after_costs < MIN_R_AFTER_COSTS
stop_vs_noise	abs(entry-stop) < 0.40*atr_14
stop_too_wide	abs(entry-stop) > 2.5*atr_14
liquidity	size vs depth/volume rule above
funding_carry	funding_est > 0.35 * risk_cash
cluster	already a paper position in same asset; or same-direction count would exceed 2; or concurrent >= MAX
daily_loss	paper day PnL <= -2% equity
weekly_loss	paper week PnL <= -5% equity
circuit	system_state.mode=halted
hist_cell_fatal	optional later: if cell n>=80 AND mean_r < -0.15 after costs — fail. If n small, do not fail (unproven ≠ forbidden at research size).

Soft gates add reasons but do not alone block unless 3+ fire: vol_ratio<0.6, ADX in [16,20], spread_bps>5 for BTC/ETH or >12 for SOL, LLM packet incomplete.

SYSTEM RULE

Hard gates cannot be overridden by the LLM, Telegram commands, or 'confidence'. /force does not exist.

10. Decision Pipeline (Exact Order)

Function pipeline.evaluate(asset, timeframe, bar_open_time). Stop at first hard fail. Always persist an idea if a detector fired; persist a scan_event if none fired.

Step	Code action	Fail
1 Data valid	integrity.ok for candles/ctx/book	NO_TRADE
2 Regime	regime.classify	NO_TRADE if UNKNOWN/PANIC/EVENT or mismatch
3 Setups	run 3 detectors	if none: scan_event only
4 Historical support	lookup cell (setup, regime primary, tf, asset)	annotate; fail only if hist_cell_fatal
5 Volatility	atr_pct band vs setup	NO_TRADE if PANIC already; HIGH_VOL shrinks size 50% if policy on — MVP: no shrink, only flag
6 Liquidity	book + dayNtlVlm vs size	NO_TRADE
7 Funding/OI	carry + crowding flag (funding_z>2.5)	NO_TRADE if carry gate; else annotate

Step	Code action	Fail
8 News/macro	calendar + news last 6h titles only	EVENT_HIGH already failed; else annotate NEWS_PRESENT
9 Conflict	opposite setups or existing paper position	NO_TRADE
10 Entry	geometry.entry	NO_TRADE if unexecutable
11 Invalidation	geometry.stop	NO_TRADE if unclear
12 Expected R	raw_R - cost_R	NO_TRADE if < MIN_R
13 Risk filters	cluster, daily/weekly, circuit	NO_TRADE
14 Hist cell	attach n, mean_r, std, last_eval	annotate unproven
15 LLM review	only if all hard gates passed	see §11; LLM cannot TAKE if it invents levels
16 Confidence	code confidence = min(regime.conf, 0.6 if n<30 else 0.7) * llm.agreement	NO_TRADE if < 0.35
17 Wait check	minutes to funding hour < 5 on 15m; incomplete extras	WAIT
18 Final	TRADE_PAPER | WAIT | NO_TRADE	prefer NO_TRADE

WAIT means: persist idea, do not open paper position, rescan next bar. Do not spam Telegram WAIT unless /verbose.

11. LLM Contract

11.1 When the LLM is called

Only if hard gates 1–13 passed. Rate limit: max 30 calls/hour and max 200/day. If over limit: decision WAIT reason=llm_budget, do not invent a TAKE.

11.2 Input packet (exact keys)

```
{
"schema": "agent.llm_packet.v1",
"idea_id": "uuid", "ts_utc": "...", "asset": "BTC", "timeframe": "1h",
"strategy_version_id": "sv_...", "prompt_version_id": "pv_...",
"mode": "paper",
"data_quality": {"ok": true, "flags": []},
"regime": {"primary":"TREND_UP","secondary":["HIGH_VOL"],"confidence":0.7},
"setup": {"id":"trend_pullback","direction":"long",
"trigger_desc":"coded","levels":{"entry":...,"stop":...,"targets":[...]}},
"features": {subset of §7.1 keys only},
"book": {"spread_bps":..., "imbalance_5":..., "notional_to_10bps":...},
"derivatives": {"funding":..., "funding_z_168":..., "oi":..., "oi_chg_24h":..., "basis_bps":...},
"costs": {"fee_rt":..., "slip_est":..., "funding_est":..., "planned_R_after_costs":...},
"portfolio": {"equity":..., "open_positions":[...], "day_pnl_pct":..., "cluster":...},
"news": [{"ts":..., "title":..., "source":...}],
"calendar": [{"name":..., "impact":..., "minutes_to":...}],
"hist_cell": {"n":..., "mean_r":..., "std_r":..., "win_rate":..., "note":"unproven|weak|ok"},
"gates_passed": ["data_valid", "..."],
```

```
"forbidden": ["do not invent levels","do not claim news caused the tape"]
}
```

11.3 System prompt constraints (frozen prompt_version)

The system prompt must include: handbook process rules in short form; packet-only levels; NO TRADE default; no indicator math; no causal news claim; output JSON only matching schema; if packet incomplete return NO_TRADE.

11.4 Output schema (strict JSON)

```
{
"schema": "agent.llm_decision.v1",
"recommendation": "TAKE" | "WAIT" | "NO_TRADE",
"agree_with_code": true,
"contradictions": ["..."],
"thesis": "<= 80 words, packet facts only",
"invalidation_restated": "must cite packet stop",
"news_causal_claim": false,
"used_invented_level": false,
"confidence": 0.0,
"what_would_change_decision": "..."
}
```

11.5 Validation and failure behavior

- Parse JSON. On schema fail, retry once with a repair prompt that includes the validator errors. Second fail: recommendation treated as NO_TRADE, reason=llm_invalid_json.
- If used_invented_level=true OR thesis mentions a price not in packet: force NO_TRADE, reason=llm_invented_level.
- If news_causal_claim=true: strip claim, set flag, do not force NO_TRADE by itself.
- If recommendation=TAKE but agree_with_code=false: NO_TRADE (structured evidence wins).
- If recommendation=NO_TRADE: final is NO_TRADE even if code gates passed.
- If recommendation=WAIT: final WAIT.
- If recommendation=TAKE and validations pass: final TRADE_PAPER.
- Timeout 45s or HTTP 5xx: WAIT reason=llm_unavailable. Do not retry more than once.
- Never call the LLM to compute size, ATR, or R.

SYSTEM RULE

Code-only baseline: persist code_decision_before_llm on every gated idea. Final decision after LLM is separate. Stats must be computable for CODE_ONLY vs CODE_PLUS_LLM on the same ideas.

12. Telegram Specification

12.1 Access

Ignore updates whose chat_id is not in TELEGRAM_ALLOWED_CHAT_IDS. Reply nothing to strangers. Rate-limit commands 10/min/chat.

12.2 Commands

Command	Behavior
/help	List commands + current mode
/status	mode, equity, day PnL, open paper count, last scan ts, integrity, circuit
/health	HL last success, DB ok, LLM last success, stale flags
/regime	Latest primary+secondary per asset per TF
/ideas [n]	Last n ideas (default 5) with decision
/idea <uuid>	Full idea summary (geometry, gates, thesis, outcome if any)
/positions	Open paper positions + MFE/MAE
/stats [setup]	Cell table: n, mean R, PF, max DD paper, CODE vs LLM split
/journal [n]	Compact journal
/version	strategy_version_id, prompt_version_id, git sha
/halt [reason]	Set halted; stop new TRADE_PAPER
/resume	Clear halt only if integrity.ok
/mode	Show mode; cannot switch to testnet in MVP
/verbose on|off	WAIT alerts on/off

SYSTEM RULE

No command may change setup parameters, risk fraction, or universe. Those require a code change + new strategy_version.

12.3 Alert format (TRADE_PAPER or important NO_TRADE only if verbose)

```
TITLE: PAPER | BTC | 1h | trend_pullback | LONG
DECISION: TRADE_PAPER | planned R after costs: 1.41
ENTRY / STOP / T1: ...
REGIME: TREND_UP + LOW_VOL | ADX 22 | ATR% 0.9
COST: fee 0.09R slip 0.08R funding 0.04R
GATES: all hard passed
HIST: n=14 meanR=n/a (unproven)
LLM: TAKE agree=true conf=0.46
THESIS: ...
ID: uuid | sv=... pv=...
```

One message per idea that becomes TRADE_PAPER. Edits: send a follow-up on fill (paper) and on close with R and class.

13. Paper Monitoring, Outcomes, MFE/MAE, Counterfactuals

13.1 Open-state machine

PENDING_ENTRY → OPEN → CLOSED. MVP uses immediate paper fill at model price when decision is TRADE_PAPER (no intra-bar limit simulation). Status OPEN until stop, target, time-stop, or halt-flatten rule.

Exit	Condition
target	high (long) or low (short) of a closed bar reaches target — conservative: use closed bar extreme, not forming
stop	low/high of closed bar breaches stop; fill = stop * adverse slip
time_stop	HOLD_BARS_DEFAULT * 2 bars elapsed without target — flatten at close
halt_flatten	circuit daily/weekly — flatten at next close
invalidation_other	reserved

13.2 MFE / MAE

```
While OPEN, each new closed bar:
long MFE_px = max(MFE_px, bar.high); MAE_px = min(MAE_px, bar.low)
MFE_R = (MFE_px - entry) / (entry - stop) [long]
MAE_R = (entry - MAE_px) / (entry - stop)
```

Persist on every monitor tick into the position row. On close, freeze.

13.3 Outcome classification (code, then optional LLM comment)

Class	Rule
target_hit	exit_reason=target
stop_hit	exit_reason=stop
time_stop	exit_reason=time_stop
halt_flatten	halt
data_problem	integrity fail during life flagged

Loss taxonomy (handbook): applied only if realized_r < 0. Default class=random_variance. Upgrade to regime_mismatch if regime flipped before stop. unexpected_news if a high-impact item timestamped after entry. stop_too_tight if MFE_R >= planned target R and MAE_R > 1.0 would have been needed. Computed numerically first.

13.4 Counterfactuals (numeric, stored on close)

- cf_delay_1bar: if entry at next bar close, would stop have hit before target? realized_r?
- cf_stop_plus_1atr: widen stop 1*atr_14 at entry time; size would have changed — report path only, not resized PnL, plus whether target hit first.
- Do not feed these to LLM until numbers exist. Do not change strategy from one row.

14. Statistics, Learning, Versioning

14.1 Evaluation cells

Cell key: (strategy_version_id, setup_id, asset, timeframe, regime_primary, llm_involved bool). Metrics: n, mean_r, std_r, median_r, win_rate, avg_win_r, avg_loss_r, profit_factor, expectancy_r, max_dd_r_sumpath, avg_mfe, avg_mae, avg_bars_held. Update daily and on-demand /stats.

SYSTEM RULE

Do not report a cell as an edge if n < 30. Language: n<30 unproven; 30–79 tentative; >=80 eligible for review. These thresholds are process thresholds, not statistical laws.

14.2 CODE_ONLY vs CODE_PLUS_LLM

For every idea that passed hard gates, store code_would_take=true. LLM may veto. Compare mean_r of positions that opened (CODE_PLUS_LLM) vs a shadow paper book that would have opened every hard-gated idea (CODE_ONLY shadow). Shadow book is first-class: table paper_positions_shadow or a flag book=shadow.

14.3 Learned rules

Nightly job proposes rules of the form: setup X AND feature_pred Y in regime Z has mean_r different from cell baseline. Write learned_rules status=proposed. Never auto-promote. Operator promotion is a future SQL/admin action; MVP only proposes and lists in /stats.

Required fields on a proposal: definition, n, mean_r, ci if computed with naive SE, regime, setup, validation_period, parent strategy_version, status.

SYSTEM RULE

A single loss never writes a learned_rule and never changes detectors.

14.4 Versioning

strategy_version_id = 'sv_' + first 12 of sha256(canonical_json(params)+git_sha). prompt_version_id = 'pv_' + first 12 of sha256(system_prompt+output_schema+model). ideas store both. Changing EMA length without a new version is a spec violation.

15. Integrity, Failures, Logging, Circuit Breakers

15.1 Integrity checks

- Candle gap: expected next open_time missing after 2x interval → flag + skip scan.
- Stale ctx/book: age > 60s at scan time → fail data_valid.
- Duplicate candle unique key: upsert.
- Implausible jump: |log(c/c_prev)| > 0.15 on BTC/ETH 15m (tune in params) → flag, do not auto-trade.
- Mark vs mid: |mark-mid|/mid > 0.01 → annotate; still usable for liq theory but paper uses last/close.
- Clock: host UTC offset vs HL time from ctx; if |skew| > 5s log warning; > 30s halt scans.
- Restart recovery: on boot, mark running job_runs as failed, reload open paper_positions, rebuild feature warmup from DB, do not double-open ideas with same (asset,tf,setup,bar_open_time).

15.2 API failure

HL 429: backoff, skip this tick. Three consecutive ctx failures: integrity.hl_down=true, halt new trades, Telegram /health alert once per 15 min. LLM down: WAIT, not TAKE. DB down: process exit nonzero so supervisor restarts; do not buffer ideas in memory as source of truth.

15.3 Circuit breakers

Breaker	Trip	Reset
daily_loss	paper day PnL <= -2% of start-of-day equity	next UTC day auto if /resume not required — MVP: require /resume
weekly_loss	week <= -5%	/resume + operator ack
integrity_halt	hl_down or clock skew or gap on all assets	auto when checks clear
manual_halt	/halt	/resume
llm_error_burst	10 invalid JSON in 30 min	auto 60 min cooloff

15.4 Logging

JSON logs to stdout. Fields: ts, level, logger, job, asset, tf, idea_id, event, duration_ms. No secrets. Rotate via docker/journald. Persist audit_log for halt/resume/commands.

16. Testing, Backtesting, Walk-Forward, Paper Mode

16.1 Automated tests (MVP must have)

- Unit: swing pivots, ATR, regime labels on fixture series, each detector on hand-built OHLCV, cost math, gate matrix, JSON schema validator, conflict rule.
- Lookahead tests: detector that peeks at bar t+1 must fail CI.
- Idempotency: pipeline twice on same bar produces one idea (unique on asset,tf,setup,bar_open_time).
- Telegram auth: unknown chat dropped.
- LLM fixture: invented level → forced NO_TRADE.
- Integrity: gap → no TRADE_PAPER.

16.2 Backtesting (research tool, not live)

scripts/eval_walkforward.py reads stored candles only (no book → use conservative slip_bps=5 and spread assumption). Must include fees + fundingHistory if present else assume 0.0000125/hour carry cost as a parameter, not as truth. Entry on close of signal bar is optimistic; report also T+1 open variant. OHLC path for stops uses conservative intra-bar rule: if both stop and target in same bar, assume stop first (pessimistic).

16.3 Walk-forward

If enough 1h bars: train windows are unused for parameter search in MVP because parameters are frozen. Walk-forward in MVP = out-of-time reporting only: compute cell metrics on rolling 30-day slices without changing params. No optimizer loop.

16.4 Paper-trading mode

Default AGENT_MODE=paper. Fills as §9–13. Equity curve in paper_equity. This is the only mode implemented.

16.5 Later: Hyperliquid testnet execution (NOT BUILT)

Separate spec addendum later. Planned shape only:

- AGENT_MODE=testnet_exec plus HL_ACCOUNT_ADDRESS, HL_AGENT_PRIVATE_KEY loaded from a file mode 0600, never from git.
- Base URL switches to testnet info/exchange endpoints per live docs (verify; do not hardcode stale hosts).
- Order state machine: submit, ack, reject, partial, cancel. No new risk if cancel path unhealthy.

- Sizes from the same geometry but clipped to testnet balances.
- Testnet fills must not be mixed into mainnet-data expectancy cells (book=testnet).
- Acceptance of testnet phase is out of scope for this MVP spec.

17. Security, Deployment, Monitoring, Recovery

17.1 Security

- Single operator chat allowlist.
- Secrets only in env or docker secrets. .env gitignored.
- No wallet keys in MVP.
- LLM packet must not include secrets.
- Disable Telegram group privacy issues by using a private chat only.
- DB listens on localhost only.
- VPS firewall: 22 and nothing else public unless Telegram webhook; prefer long polling to avoid opening 443.

17.2 Deployment (cheap)

docker-compose: service worker + service postgres:16. 1 vCPU / 2 GB box is enough. Bind mounts: ./data/postgres, ./logs. Restart: unless-stopped. No k8s. No managed cloud DB required. Image: python:3.11-slim + deps from pyproject.

17.3 Monitoring

Telegram /health on a 15-min watchdog job if last successful ctx_poll > 2 min. Optional: Uptime Kuma on same box pinging a local /healthz HTTP on localhost:8080. No Datadog.

17.4 Recovery after restart

- Read system_state.mode (do not default to trading if it was halted).
- Re-open cursors from DB: last candle per asset/TF, open paper_positions, peak equity.
- Recompute features from candles; do not require RAM state.
- Unique idea key prevents duplicate paper entries for the same bar after crash mid-pipeline.
- If crash during OPEN fill insert, transaction must be all-or-nothing (idea decision + position insert one txn).

18. MVP Implementation Order

Implement in this sequence. Do not start LLM or Telegram alerts before step 6 works.

#	Deliverable	Done when
1	Repo, config, docker-compose, migrations	DB up, schema applied
2	hl_client + candle/ctx/book ingest + backfill	BTC 1h candles persist and update
3	integrity + scheduler skeleton	gaps detected; jobs idempotent
4	features + regime on fixtures + live bars	feature_snapshots rows appear
5	three detectors + geometry + costs + gates	ideas written with NO_TRADE majority
6	paper monitor MFE/MAE exits	a synthetic idea closes with R

#	Deliverable	Done when
7	Telegram commands /status /ideas /halt	allowlist works
8	LLM packet + schema validation + veto logic	invented level test passes
9	alerts for TRADE_PAPER	one formatted message
10	stats views + CODE vs LLM shadow	/stats returns numbers
11	learned_rule proposer (status=proposed only)	row inserted nightly if n allows
12	tests in CI locally (pytest)	lookahead and idempotency tests green

19. MVP Acceptance Criteria

The MVP is accepted when all of the following are true on a running box for 7 consecutive days:

- Universe is only BTC, ETH, SOL and TFs 15m/1h/4h.
- AGENT_MODE=paper; no signed HL orders appear in any log.
- Every closed 1h bar produces either scan_events or ideas; no silent skip unless halted/integrity.
- Unique (asset,tf,setup,bar_open_time) — zero duplicates.
- Hard gate failures never result in TRADE_PAPER.
- LLM cannot open a trade when gates failed (LLM is not even called).
- LLM invented-level fixture forced NO_TRADE.
- At least one TRADE_PAPER lifecycle has MFE, MAE, realized_R, fees, funding fields populated.
- NO_TRADE ideas are stored, not only takes.
- /halt prevents new TRADE_PAPER; /resume requires integrity.ok.
- Restart does not duplicate positions.
- Handbook file is present and unused as executable logic.
- strategy_version_id and prompt_version_id populated on every idea.
- /stats can show n and mean_r by setup even if n is small and labeled unproven.
- Cost of LLM+VPS remains compatible with the $20 target at default rate limits.

19.1 Explicitly NOT built in MVP

- Live mainnet execution, testnet execution, wallet signing.
- Additional assets or timeframes (5m, 1D).
- Additional setups beyond the three hypotheses.
- Autopromotion of learned rules or parameter search.
- Vector DB, RAG over the handbook (optional later; excerpts may be static in the prompt only).
- Kubernetes, Kafka, Redis (Postgres is enough; if a cache is tempting, do not add it).
- Web UI, dashboards beyond Telegram.
- Multi-user / SaaS.
- Predictive ML models, RL, fine-tuning.
- Cross-exchange arb, funding arb, market making.
- L3 data, liquidation heatmaps as signals.
- News-causation engine.
- Mobile apps.

• Tax lots.

20. State Machines and API Contracts (Internal)

20.1 Idea state

```
DETECTED → GATED_FAIL → NO_TRADE
DETECTED → GATED_PASS → LLM_PENDING → LLM_FAIL → NO_TRADE|WAIT
GATED_PASS → LLM_OK_TAKE → TRADE_PAPER → POSITION_OPEN → POSITION_CLOSED
GATED_PASS → LLM_OK_WAIT → WAIT (terminal for that bar)
GATED_PASS → LLM_OK_NOTRADE → NO_TRADE
```

20.2 Unique keys

```
ideas_unique: (asset, timeframe, setup_id, bar_open_time, strategy_version_id)
```

20.3 Healthz (localhost)

```
GET /healthz → 200 {db:true, hl_age_s:n, mode, halted, open_positions}
503 if db false or hl_age_s > 120
```

20.4 Dependencies (pin in pyproject)

python 3.11, httpx, psycopg[binary], sqlalchemy or psycopg only (choose psycopg + raw SQL for simplicity), apscheduler, pandas, numpy, pandas-ta, pydantic v2 (packet validation), python-telegram-bot, structlog or logging, pytest. No langchain. No unofficial HL trading SDK required for MVP data; raw HTTP is enough. Official SDK optional only as a wrapper around the same info calls.

20.5 Coding agent constraints

SYSTEM RULE

If a choice is not specified here, pick the simpler option and document it in README. Do not add a second LLM. Do not add assets. Do not weaken hard gates. Do not implement /force.

21. Alignment with the Handbook

Handbook rule	Where implemented
NO TRADE default	pipeline step 18; detectors may return null
LLM does not compute indicators	features.py only; packet precomputed
LLM does not invent levels	validator used_invented_level + price allowlist
News ≠ causation	news_causal_claim forced false path
No double-counting indicators	feature subset is small; no extra 'confluence' score
Costs in EV	planned_R_after_costs gate
Versioning + frozen eval	strategy_version frozen flag
Single loss no rewrite	learning.py proposals only
Venue docs override	hl_client raw fields; no hardcoded mark formula

Handbook rule	Where implemented
Three setups as hypotheses	setups/* + hist_cell unproven language

SYSTEM RULE

End of specification. Implementation may begin at Part 18 step 1. Any deviation that adds execution, assets, setups, or autonomous rewriting requires a new spec version, not a silent commit.